# Detailed Evaluation of Ecosystem Processes (DEEP) Field Worksheet

WY SQT Adapted | version 1
Wyoming SQT

**Reach ID: _____ Reach Length: _____ Date: _____ Assessor(s): _____ Coordinates: _____**

**RECORDING INSTRUCTIONS**

Record the measured value of each metric in its units. Metrics measured in the field are marked F and metrics answered from the desk are marked D. A value printed in the Value cell was already answered from the desk and carries its source. Enter the values in DEEP, which reads each one on the metric's reference curve to give an index from 0 to 1. The function score is the mean index of the function's metrics times 15 and reads Functioning (15 to 11), Functioning At-Risk (10 to 6), or Non-Functioning (5 to 0). A metric that does not apply is marked NA and drops out of the score. Where a metric lists curve sets, mark the set that fits the reach.

| Function | Metrics | Value | Index |
|---|---|---|---|
| **HYDROLOGY FUNCTIONS** | | | |
| **Catchment hydrology:** Runoff and infiltration sustain natural flow regime, carry appropriate sediment and nutrients from uplands, and reliably cue spawning/migration of aquatic life. | **Land Use Coefficient** D: Context: Watershed Method: Desktop (GIS) | | |
| **Score:** | **Notes/Other Metrics:** | | |
| **Reach inflow:** Quantity + quality of inflow (tributaries, ditches, and pipes) does not provide harmful peaks or pollution, and supports diverse habitat conditions. | **Concentrated Flow Points** D: Context: Channel Method: Field (with desktop pre-screen) | | |
| **Score:** | **Notes/Other Metrics:** | | |
| **Streamflow regime:** Integrates the range of "typical" flows experienced by other processes. Examines the degree to which upstream infrastructure or land uses have fundamentally altered flow regimes (e.g., hydropeaking, dams, withdrawals). | **Flow Alteration Module** D: Context: Watershed Method: Desktop (hydrology/flow analysis) | | |
| **Score:** | **Notes/Other Metrics:** | | |
| **HYDRAULICS FUNCTIONS** | | | |
| **Low flow and baseflow dynamics:** Ensures habitat availability and water quality during low water levels, and indicates non-storm conditions experienced the majority of the time. | **Average Depth** F: Context: Channel Method: Field<br>Curve set: [ ] Tier I, W < 20' [ ] Tier I, W >20' [ ] Tier II | | |
| **Score:** | **Notes/Other Metrics:** | | |

| | | | |
|---|---|---|---|
| **Floodplain connectivity:** Frequent overbank flows reduce flood peaks, support riparian vegetation, create off-channel refugia, and extend nutrient processing time. | **Entrenchment Ratio (ER)** (ER) D: Context: Channel/Floodplain Method: Field (survey) or Desktop (LiDAR cross sections)<br>Curve set: [ ] Cb Stream Types [ ] C Stream Types [ ] E Stream Types [ ] A, B, Ba, and Bc Stream Types | | |
| **Score:** | **Notes/Other Metrics:** | | |
| **GEOMORPHOLOGY FUNCTIONS** | | | |
| **Channel evolution:** Long-term changes in channel size and slope maintain dynamic equilibrium and diverse habitats while interacting with the floodplain. | **Width/Depth Ratio State** F: Context: Channel Method: Field (survey) | | |
| **Score:** | **Notes/Other Metrics:** | | |
| **Channel and floodplain dynamics:** Examines the role of bank processes and erosion in channel change. Observes patterns in sinuosity and curvature affecting habitat complexity. | **Bank Height Ratio (BHR)** (BHR) D: Context: Channel/Riparian Method: Field | | |
| | **Greenline Stability Rating** F: Context: Channel/Riparian Method: Field | | |
| | **Percent Side Channels** F: Context: Varies Method: See state SQT protocol<br>Curve set: [ ] Unconfined Alluvial [ ] Confined Alluvial | | |
| | **Percent Streambank Armoring** F: Context: Channel/Riparian Method: Field | | |
| | **Percent Streambank Erosion** F: Context: Channel/Riparian Method: Field | | |
| **Score:** | **Notes/Other Metrics:** | | |
| **Bed composition and large wood:** Supports aquatic habitats through streambed material and bedform dynamics. Enhances habitat complexity and streambank stability. | **Percent Fines < 2mm** F: Context: Channel Method: Field | | |
| **Score:** | **Notes/Other Metrics:** | | |
| **PHYSICOCHEMISTRY FUNCTIONS** | | | |
| **Light & thermal regime:** Shade, solar input, and groundwater keep temperature and light within natural ranges for stream metabolism and dissolved oxygen. | **Maximum Weekly Average Temperature (MWAT)** (MWAT) D: Context: Channel Method: Field (logger) and/or Desktop (data compilation)<br>Curve set: [ ] Tier I (Cold) [ ] Tier II (Cold-Cool) [ ] Tier III (Cool) [ ] Tier IV (Cool-Warm) [ ] Tier V (Warm) | | |
| **Score:** | **Notes/Other Metrics:** | | |

| | | | |
|---|---|---|---|
| **Carbon processing:** Organic matter is captured and broken down, fueling food webs, balancing production/ respiration, moderating pH/ redox, and supplying nutrients. | **Chlorophyll ?** F: Context: Channel Method: Field (sampling/assay)<br>Curve set: [ ] Basins and Plains [ ] Mountains | | |
| | **Herbaceous Vegetation Cover** D: Context: Riparian/Floodplain Method: Field and/or Desktop (imagery/GIS) | | |
| | **Percent Native Cover** D: Context: Riparian/Floodplain Method: Field and/or Desktop (imagery/GIS) | | |
| | **Riparian Extent** D: Context: Riparian/Floodplain Method: Field and/or Desktop (imagery/GIS)<br>Curve set: [ ] Unconfined Alluvial Valleys [ ] Confined Alluvial, Colluvial/V-Shaped, or Bedrock Valleys | | |
| | **Woody Vegetation Cover** D: Context: Riparian/Floodplain Method: Field and/or Desktop (imagery/GIS)<br>Curve set: [ ] Mountains and Basins [ ] Plains | | |
| **Score:** | **Notes/Other Metrics:** | | |
| **BIOLOGY FUNCTIONS** | | | |
| **Habitat provision:** Channel and floodplain structure supply depth, velocity, substrate diversity, and vegetation to support native organisms through all life stages. | **LWD Index** F: Context: Channel Method: Field | | |
| | **LWD Piece Count** F: Context: Channel Method: Field | | |
| | **Percent Riffle** F: Context: Channel Method: Field<br>Curve set: [ ] Streams < 3% slope [ ] Streams < 3% slope / Overall [ ] Volcanics Mountains and Valleys Bioregion | | |
| | **Pool Depth Ratio** F: Context: Channel Method: Field | | |
| | **Pool Spacing Ratio** F: Context: Channel Method: Field<br>Curve set: [ ] B and Ba Stream Types [ ] Bc Stream Types [ ] C Stream Types [ ] Cb Stream Types [ ] E Stream Types | | |
| **Score:** | **Notes/Other Metrics:** | | |
| **Population support:** Physical and chemical conditions enable spawning, juvenile growth, migration, and stress survival of target taxa. | **Game Species Biomass** F: Context: Channel Method: Field (fish survey)<br>Curve set: [ ] Blue Ribbon and non-trout game fish [ ] Red Ribbon [ ] Yellow/Green Ribbon | | |
| **Score:** | **Notes/Other Metrics:** | | |
| **Community dynamics:** Species interactions sustain biodiversity, restrain invasives, and build resilience to disturbance. | **Native Fish Species Richness** D: Context: Channel Method: Field (fish survey) and/or Desktop (existing data) | | |
| **Score:** | **Notes/Other Metrics:** | | |